

Abhinay Kalkhanday

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[Portfolio](#)

[LinkedIn](#)

[GitHub](#)

EDUCATION

Manipal University Jaipur

B.Tech in Computer Science (IoT & Intelligent Systems)

Jaipur, Rajasthan

Exp. Graduation: 2026

EXPERIENCE

Matrix Education

Backend Developer

Jaipur, Rajasthan

Jan 2026 – Present

- Improved performance across **12 production websites**, reducing load time from **45s to 5.6s**, **LCP from 9.6s to 0.9s**, and **TTFB from 4.0s to 0.3s**.
- Supported platforms serving **31k monthly visitors** and **388k requests/month**, optimizing CDN delivery and caching using Cloudflare.
- Designed reusable CMS architectures and browser automation workflows using Playwright and Express.js across multiple production websites.
- Investigated and mitigated a production SEO spam incident affecting a live website by auditing database records, hardening deployments, and integrating automated dependency vulnerability monitoring.

Intern ATS & SCADA Systems

Jaipur Metro Rail Corporation

Jaipur, Rajasthan

May 2024 – June 2024

- Analyzed complex **ATS & SCADA architectures** to author a technical training suite, subsequently adopted as the standard onboarding material for new interns.

PROJECTS

Content Operations Automation Platform | [GitHub](#)

Node.js, Express.js, Playwright, SQLite, PM2

- Designed a stateful job orchestration platform for long-running browser workloads using persistent queues, priority scheduling, and durable job storage.
- Implemented automatic crash recovery using a finite state machine supporting **8 worker states**, **7 failure categories**, and up to **5 recovery attempts**, restoring service within **60 seconds** after browser failures.
- Automated content maintenance workflows by processing stale or low-quality articles, identifying content issues, generating updated drafts, and routing rewritten posts to a human review queue before publication.

Judicial Transparency Delay Analytics Platform | [GitHub](#)

FastAPI, PostgreSQL, Redis, Celery, Docker, Next.js

- Designed a judicial transparency platform that aggregates and analyzes public court data from **4 Indian judicial sources** to identify delay patterns, adjournments, and case lifecycle trends.
- Built resilient ingestion pipelines with schema validation, anomaly detection, audit logging, and automated recovery to handle failures and source changes without data loss.
- Developed distributed background workflows using **29 Celery tasks**, Redis, and scheduled retraining pipelines for ingestion, monitoring, and machine learning operations.
- Implemented machine learning models for case duration prediction, hearing outcome classification, and delay severity detection using historical judicial data.

Electra-Core: Intelligent Voter Roll Extraction Platform | [GitHub](#)

OpenCV, Tesseract OCR, Streamlit

- Designed a modular document intelligence architecture for extracting structured voter records from degraded voter roll PDFs using computer vision and OCR pipelines.
- Implemented a strategy-based extraction engine using **4 independent extraction strategies** with automatic fallback handling for varying document layouts and scan quality.
- Built a quality-gated validation pipeline using **5 validation rules** to detect extraction failures and automatically route low-confidence records for manual review.
- Developed a Human-In-The-Loop review system using Streamlit for correction workflows, audit reporting, and structured Excel export generation.
- Processed 26-page voter roll documents containing up to **780 voter records** in approximately **30–75 seconds** while maintaining end-to-end traceability of extraction decisions.

TECHNICAL SKILLS

Languages

Java, Python, JavaScript, PHP, SQL

Backend

FastAPI, Express.js, Node.js, REST APIs

Databases

PostgreSQL, Redis, SQLite, MySQL, MongoDB

Infrastructure

Docker, Git, GitHub Actions, Cloudflare, Postman

Automation

Playwright, Browser Automation, Job Queues, Workflow Orchestration

ML/AI

Scikit-learn, spaCy, OpenCV, Tesseract OCR

Frontend

HTML5, CSS3, Vanilla JavaScript